POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2020
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define Signal Analysis and Signal Processing. What are the advantages of Digital signal processing over Analog signal processing? 8
   b) Define elementary signals. State and prove Convolution and time shifting property of Discrete time Fourier transform. 7
2. a) Find the response of a LTI system with input x(n) = {9,8,7} to an impulse response h(n) = {5, -4, **8**}, where bold letter denotes origin. Use graphical approach. Also verify your answer. 8
   b) Define Z–transform and ROC. Determine the Z-transform and plot the ROC for the given signal 7

$$x(n) = (-1)^n Cos\left(\frac{\pi}{4} n\right) u(n)$$

3. a) Why we need DFT when we have DTFT? Find the circular convolution of the sequence $x_1(n)=\{9,8,7,6\}$ and $x_2(n)=\{6,4,3,2\}$ using convolution sum. 8
   b) How efficient is FFT? Determine 8-point DFT of the sequence $x(n) = \{1, 0, 2, 0, 3, 0, 1, 1\}$ using Decimation in Frequency Radix-2 Butterfly structure (DIFFFT). 7
4. a) Determine the direct form realizations of the following difference equation. 8

$$2y(n) + 3y(n-1) + 5y(n-2) = x(n) + 2x(n-3)$$

   Compare FIR and IIR.
   b) Compute the lattice coefficients and draw the lattice structure of following FIR system 7

$$H(z) = 1 + 2z^{-1} + z^{-2}$$

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5. a) How can you design FIR filter using rectangular window? Explain.

   b) Design a lowpass FIR with 7 coefficients for the following specifications.

   Passband Frequency edge = 300 Hz

   Sampling Frequency = 1 KHz

   Use Hanning window for your design.

6. a) Compare and Contrast Impulse Invariance Method and Bilinear Transformation Method of designing IIR filter.

   b) Design a lowpass IIR filter to meet the following specifications

   - Passband attenuation : 1 dB
   - Passband frequency: 1.2 KHz
   - Stopband attenuation: 40 dB
   - Stopband frequency: 2.5 KHz
   - Sampling Frequency: 10 KHz

7. Write short notes on: **(Any two)**

   a) Kaiser window

   b) Causality and stability of LTI system

   c) Sampling of CT signal

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